

Unreal Engine Projection Mapping SDK - Quick Start Guide

Welcome to the Projection Mapping SDK for Unreal Engine 5! This plugin leverages our high-performance, multithreaded native C++ SDK to provide blazing fast projection mapping right inside Unreal.

Installation

1. Copy this entire ProjectionMapping folder into your Unreal project's Plugins/ directory (create the directory if it doesn't exist).
2. The folder structure should look like:
YourProject/Plugins/ProjectionMapping/ProjectionMapping.uplugin
3. Launch your Unreal Project.
4. When prompted, click Yes to rebuild the plugin modules if necessary.
5. Ensure the plugin is enabled under Edit > Plugins > Projection Mapping.

Getting Started

Using Blueprints

The SDK is fully exposed to Blueprints! 1. Right click in any Blueprint Event Graph and search for Projection Mapping. 2. Use the Create Warp Node node to initialize a new projector. 3. Use the Apply Deformation Field node to warp a Static Mesh or Procedural Mesh dynamically.

Using C++

To use the SDK from your own C++ game modules: 1. Open your project's Build.cs file (e.g. YourProject.Build.cs). 2. Add "ProjectionMappingSDK" to your PublicDependencyModuleNames. 3. In your code, you can include the C-API headers:

```
```cpp
```

```
include "PMSDK/C_API/WarpAPI.h"
```

```
include "PMSDK/C_API/GeometryAPI.h"
```

```
void AMyActor::StartMapping() { // Call the native C-API directly! pmsdk_warp_node_h warpNode =
pmsdk_warp_node_create("MyProjector");
```

```
// ... setup and projection logic ...
```

```
}```
```

## Troubleshooting

If you encounter linker errors regarding ProjectionMappingSDK.dll, ensure that you copied the bin/ folder correctly when you packaged the release. The Unreal plugin dynamically loads the native DLL from Source/ThirdParty/ProjectionMappingSDK/bin/Release/ProjectionMappingSDK.dll.